$\mu = 5000$
$\sigma = 50$

Total number of coin tosses with heads
$= X_n$

$$P(X_n \geq \mu + 4\sigma) = P(X_n \geq 5200)$$
$$\leq \frac{\sigma^2}{\sigma^2 + 4^2\sigma^2} = \frac{1}{17}$$

by Chebyshev's inequality

Via CLT, this prob. is <= $10^{-4} = \frac{1}{10,000}$

$$\int_x^{\infty} \left(e^{-t^2/2}\right) t\, dt \qquad t^2 = y, \quad 2t\, dt = dy$$

$$= \int_{x^2}^{\infty} e^{-y/2} \frac{dy}{2} \qquad t = x, \; y = x^2; \quad t = \infty, \; y = \infty$$

$$= \frac{1}{\cancel{2}} \left(\frac{e^{-y/2}}{-1/\cancel{2}}\right)_{x^2}^{\infty} = -\left(0 - e^{-x^2/2}\right)$$
$$= e^{-x^2/2}$$

Given ~~samples~~ values $x_1 \ldots x_n$ of a random variable X, the variance of X is calculated as follows

$$\mathrm{Var}(X) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

empirical / sample variance

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$